Fractional, Circular, and Defective Coloring of Series-Parallel Graphs

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Abstract

In this note we consider colorings of series-parallel graphs. Specifically, we provide bounds on their fractional and circular chromatic numbers and the defective version of these parameters. The main result is that the fractional chromatic number of any series-parallel graph of odd girth k is exactly $2k/(k-1)$.

1 Introduction

In this paper we consider colorings of series-parallel graphs. In particular, we show that the fractional chromatic number of a series-parallel graph is determined by the ***odd girth*** (length of shortest odd cycle) of the graph. By ***series-parallel*** we will mean a graph without a K_4-minor (as used for example in [6]).

We will consider the following colorings. A (k,q)***-fractional coloring*** [14] is an assignment of q colors to each vertex, where the colors are drawn from a palette of k colors, such that adjacent vertices receive disjoint q-sets. A (k,q)-***circular coloring*** [16] (originally called star coloring) is an assignment of one color to each vertex, where the colors are drawn from $\mathbb{Z}_k$, such that adjacent vertices receive colors that are at least $q \pmod{k}$ apart. It is well known that if there is a (k,q)-circular coloring, then there is a (k,q)-fractional coloring. For a survey of circular colorings, see Zhu [20]. The ***fractional chromatic number*** $\chi_f(G)$ and the ***circular chromatic number*** $\chi_c(G)$ of a graph G are defined as the infimum of k/q taken over all fractional colorings and all circular colorings of G respectively. It is well known that the infimum is achieved; that is, one can replace infimum by minimum.

A ***d-defective coloring*** [3] (also called a d-improper coloring) is an assignment of one color to each vertex such that every vertex has at most d neighbors of the same color. Equivalently, a graph has a d-defective coloring if one can remove the edges of a subgraph of maximum degree d such that the result is a proper coloring. Similarly, a ***d-defective (k,q)-fractional coloring*** [15] is an assignment of q colors to each vertex, where the colors are drawn from a palette of k colors, such that every vertex v is adjacent to at most d vertices u where the q-set of v overlaps the q-set of u. A ***d-defective (k,q)-circular coloring*** [9] is an assignment of one color to each vertex, where the colors are drawn from $\mathbb{Z}_k$, such that every vertex v is adjacent to at most d vertices u where the difference between the color of v and the color of u is less than $q \pmod k$. Since the number of subgraphs of a graph is finite, it follows similarly that one can define the ***d-defective fractional chromatic number*** and the ***d-defective circular chromatic number*** as the minimum of k/q taken over all d-defective (k,q)-fractional colorings and all d-defective (k,q)-circular colorings of a graph respectively. (Defective circular colorings have been generalized by Mihók et al. [11] by considering alternative requirements on the graph induced by the improper edges.)

It is well known that series-parallel graphs are 3-colorable. Thus, if such a graph has a triangle, then its fractional and circular chromatic number are 3. Hell and Zhu [5] showed that a triangle-free series-parallel graph has circular chromatic number at most 8/3, and that this is best possible because of the graph of Figure 1. Pan and Zhu provided bounds for higher girth in [12], and proved that their bounds are best possible in [13].

Outerplanar graphs form a subclass of the series-parallel graphs. The results simplify for such graphs. Klostermeyer and Zhang [10] (and later Kemnitz and Wellmann [7]) observed that every outerplanar graph of odd girth k has circular chromatic number (and fractional chromatic number) exactly $2k/(k-1)$. This was extended by Wang et al. [17] who showed that the same result holds for circular list colorings of outerplanar graphs. The defective choosability of outerplanar and series-parallel graphs was also studied by Woodall [19].

We proceed as follows. In Section 2 we show that the fractional chromatic number of any series-parallel graph of odd girth k is exactly $2k/(k-1)$. In

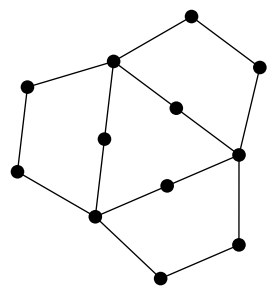

Figure 1: A series-parallel graph with circular chromatic number 8/3

Section 3 we first note that a series-parallel graph of girth 5 is 2-colorable with defect 1, and then provide constructions that show that in many cases the upper bound for the defective version of the parameter is the same as the upper bound for the ordinary fractional or circular chromatic numbers. In Section 4, we propose some related questions.

2 Proper Fractional Colorings

Wang and Yu [18] (assuming a typo in the paper) conjectured that the fractional chromatic number of any series-parallel graph of odd girth at least k is at most $2k/(k-1)$. The main goal of this section is to show that the fractional chromatic number of any series-parallel graph of odd girth k is exactly $2k/(k-1)$, which proves their conjecture.

2.1 Combining intervals

We need the following definitions and notations. Fix k to be an odd integer; say $k = 2\ell+1$. Fix a palette of k colors, and let $\mathcal{L}$ be the set of all subsets of ℓ colors. Given integers i and j such that $0 \leq i, j \leq \ell$, define $i \oplus j$ as the set of all $|S_1 \cap S_2|$ such that $S_1, S_2, S_3 \in \mathcal{L}$ with $|S_1 \cap S_3| = i$ and $|S_2 \cap S_3| = j$. Given integers a and b, let $[a, b]$ denote the set of consecutive integers $\{a, a+1, \ldots, b\}$, and call it an ***integer interval***.

The proof of our main result is based on the following lemma.

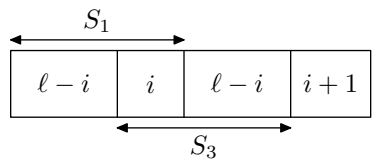

Figure 2: Computing $i \oplus j$

Lemma 1 *(a) $i \oplus j$ is the nonempty integer interval* $[\max(\ell - i - j - 1, i + j - \ell), \min(\ell - i + j, \ell + i - j)]$.
(b) Given any integer intervals I_1 and I_2, the set $I_1 \oplus I_2 = \{\, i \oplus j : i \in I_1$ and $j \in I_2 \,\}$ is an integer interval.

Proof. (a) It is easy to verify that $\max(\ell - i - j - 1, i + j - \ell) \leq \min(\ell - i + j, \ell + i - j)$, and so the above interval is nonempty. Suppose that $S_1, S_3 \in \mathcal{L}$ with $|S_1 \cap S_3| = i$. Note that there are $k + i - 2\ell = i + 1$ elements outside $S_1 \cup S_3$. See Figure 2.

To maximize $|S_1 \cap S_2|$, we take j elements from S_3 using S_1 as much as possible, and then $\ell - j$ elements outside S_3 using S_1 as much as possible. The overlap $|S_1 \cap S_2|$ is $\min(i, j) + \min(\ell - j, \ell - i)$, which simplifies to $\min(\ell - i + j, \ell + i - j)$. To minimize $|S_1 \cap S_2|$, we take j elements from S_3 avoiding S_1 as much as possible, and then $\ell - j$ elements outside S_3 avoiding S_1 as much as possible. The overlap $|S_1 \cap S_2|$ is $\max(j - (\ell - i), 0) + \max(\ell - j - (i + 1), 0)$ which simplifies to $\max(\ell - i - j - 1, i + j - \ell)$, since $(j - (\ell - i)) + (\ell - j - (i + 1)) = -1$ and therefore at least one of $j - (\ell - i)$ and $\ell - j - (i + 1)$ is nonnegative.

To complete the proof, note that we can get any value between the two extremes, by choosing differently.

(b) This follows from noting that the upper and lower limits of $i \oplus j$ change by at most 1 when we change either i or j by 1. $\square$

2.2 Coloring two-terminal series-parallel graphs

A ***two-terminal series-parallel graph*** is a graph with two distinguished vertices built recursively from the complete graph K_2 using some combination of the parallel and series joins. It is well known that every block of (what we call)

a series-parallel graph is a two-terminal series-parallel graph for some choice of distinguished vertices.

For a two-terminal series-parallel graph G, let $o(G)$ denote the length of the shortest odd path between the two terminals of G if such a path exists, and let $o(G) = \infty$ otherwise. Similarly, let $e(G)$ denote the length of the shortest even path between the two terminals of G if such a path exists, and let $e(G) = \infty$ otherwise. Let $I_\ell(G) = [\ell - e(G)/2, (o(G) - 1)/2] \cap [0, \ell]$. For any two-terminal series-parallel graph G with odd girth at least k, clearly $o(G) + e(G) \geq k$, and hence $\ell - e(G)/2 \leq (o(G) - 1)/2$. Therefore, $I_\ell(G)$ is nonempty for such a graph.

Theorem 2 *Let G be a two-terminal series-parallel graph with odd girth at least k, where $k = 2\ell + 1$. Then there is a (k, ℓ)-fractional coloring of G. Furthermore, the color sets for the two terminals of G can be specified as any pair (S_1, S_2) such that $|S_1 \cap S_2| \in I_\ell(G)$.*

Proof. We prove the theorem by induction. The base case is $G = K_2$. Here $o(G) = 1$ and $e(G) = \infty$, so $I_\ell(G) = \{0\}$. Choosing disjoint color sets S_1 and S_2 for the two terminals yields the requisite coloring.

Suppose G is obtained from graphs G_1 and G_2 by the parallel operation. Let color sets S_1 and S_2 with $|S_1 \cap S_2| \in I_\ell(G)$ be specified for the two terminals of G. Note that $\ell - e(G_j)/2 \leq \max(\ell - e(G)/2, 0) \leq |S_1 \cap S_2| \leq \min((o(G) - 1)/2, \ell) \leq (o(G_j) - 1)/2$ for $j = 1, 2$; therefore $|S_1 \cap S_2| \in I_\ell(G_1) \cap I_\ell(G_2)$. By the inductive hypothesis, graphs G_1 and G_2 have the desired coloring with S_1 and S_2 at their terminals. This yields the requisite coloring of G.

Suppose G is obtained from graphs G_1 and G_2 by the series operation. Let color sets S_1 and S_2 with $|S_1 \cap S_2| \in I_\ell(G)$ be specified for the two terminals of G. To complete the proof by induction, we need to show that we can find a color set S_3 with $|S_1 \cap S_3| \in I_\ell(G_1)$ and $|S_2 \cap S_3| \in I_\ell(G_2)$, since then we can color G_1 with S_1 and S_3 at its terminals and G_2 with S_3 and S_2 at its terminals to obtain the requisite coloring. This means that $|S_1 \cap S_2| \in |S_1 \cap S_3| \oplus |S_2 \cap S_3|$. That is, it suffices to show that $I_\ell(G) \subseteq I_\ell(G_1) \oplus I_\ell(G_2)$. By Lemma 1, it suffices to show that the extrema of $I_\ell(G)$ are contained in $I_\ell(G_1) \oplus I_\ell(G_2)$.

Consider the upper limit of $I_\ell(G)$. Assume first that $(o(G) - 1)/2 \geq \ell$, so that the upper limit of $I_\ell(G)$ is ℓ. Since $o(G) = \min(o(G_1) + e(G_2), e(G_1) + o(G_2))$,

we have $(o(G_1)-1)/2 \geq \ell - e(G_2)/2$ and $(o(G_2)-1)/2 \geq \ell - e(G_1)/2$. Therefore, $x = \max(0, \ell - \min(e(G_1), e(G_2))/2) \in I_\ell(G_1) \cap I_\ell(G_2)$. By Lemma 1, we have $\ell \in x \oplus x$; that is, $\ell \in I_\ell(G_1) \oplus I_\ell(G_2)$.

Assume second that $(o(G)-1)/2 < \ell$, so that the upper limit of $I_\ell(G)$ is $(o(G)-1)/2$. Without loss of generality we may assume $o(G) = o(G_1) + e(G_2)$. Then we have $(o(G_1)-1)/2 < \ell - e(G_2)/2$. Take $i = (o(G_1)-1)/2$ and $j = \ell - e(G_2)/2$; then by Lemma 1, it follows that $(o(G)-1)/2 = \ell + i - j \in i \oplus j \subseteq I_\ell(G_1) \oplus I_\ell(G_2)$.

Consider the lower limit of $I_\ell(G)$. Assume first that $\ell - e(G)/2 < 0$. Then, define $\ell - I_\ell(G_1)$ to be the set $\{\, j : j = \ell - i, i \in I_\ell(G_1) \,\}$. Note that both $\ell - I_\ell(G_1)$ and $I_\ell(G_2)$ are nonempty, and $\ell - I_\ell(G_1) = [\ell + (1 - o(G_1))/2, e(G_1)/2] \cap [0, \ell]$. It is easy to verify that $\ell - I_\ell(G_1)$ and $I_\ell(G_2)$ intersect; say containing integer j. Hence, $0 = (\ell - j) + j - \ell \in (\ell - j) \oplus j \subseteq I_\ell(G_1) \oplus I_\ell(G_2)$.

So suppose $\ell - e(G)/2 \geq 0$. Without loss of generality we may assume $e(G) = o(G_1) + o(G_2)$. Then take $i = (o(G_1)-1)/2$ and $j = (o(G_2)-1)/2$, and by Lemma 1, we have $\ell - e(G)/2 = \ell - i - j - 1 \in i \oplus j \subseteq I_\ell(G_1) \oplus I_\ell(G_2)$. $\square$

Our main result follows from Theorem 2.

Theorem 3 *If G is a series-parallel graph of odd girth k then $\chi_f(G) = 2k/(k-1)$.*

Proof. The expression $2k/(k-1)$ is a lower bound, since that is the fractional chromatic number of the k-cycle. The upper bound follows from Theorem 2 and the fact that the fractional chromatic number of a graph is the maximum of the fractional chromatic numbers of its blocks. $\square$

In particular, Theorem 3 shows that the fractional chromatic number of a series-parallel graph is polynomial-time computable.

3 Defective Colorings

3.1 Girth 5

We observe below (the probably known fact) that for all d there is a triangle-free series-parallel graph whose d-defective chromatic number is 3. Indeed, we note in Theorem 7 that the same is true for fractional chromatic number. However, for girth 5 the situation changes.

We will need the following observation from [1]:

Observation 4 *[1] A cyclic series-parallel graph G with girth g contains a path with $\lfloor (g-1)/2 \rfloor$ vertices each with degree 2 in G.*

Theorem 5 *A series-parallel graph G of girth 5 is 1-defective 2-colorable.*

Proof. The graph G either contains a vertex of degree 1 (in which case induction is immediate), or is cyclic and therefore by the above observation, contains two adjacent vertices of degree 2, say x and y. Apply the induction hypothesis to $G - \{x, y\}$. Then color x the opposite color to its neighbor in $G - \{x, y\}$ and similarly with y. $\square$

We note that Borodin et al. [1] considered the case where the defect condition is different for each color. A $[d_1, \ldots, d_k]$-coloring is a k-coloring of the vertices such that for each i, the vertices of color i induce a graph of maximum degree at most d_i. They showed that a series-parallel graph of girth 7 has a $[1, 0]$-coloring, and this is best possible; indeed that for all k there is a series-parallel graph of girth 6 that does not have a $[k, 0]$-coloring.

3.2 Defective Fractional and Circular Colorings

We start with a construction. We use the following notations. For disjoint two-terminal series-parallel graphs G_1 and G_2 with terminal pairs (l_1, r_1) and (l_2, r_2) respectively, we use $G_1//G_2$ to denote the parallel composition of G_1 and G_2 and $G_1 \bullet G_2$ to denote the series composition of G_1 and G_2 respectively. That is, $G_1//G_2$ is obtained from the union of G_1 and G_2 by identifying l_1 and l_2 into a

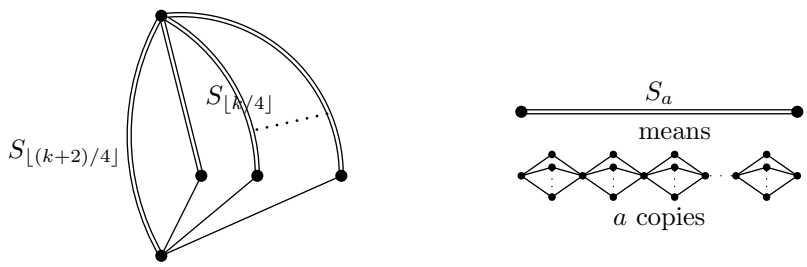

Figure 3: A series-parallel graph G_k ($k \geq 5$) with given odd girth and large d-defective fractional chromatic number

single vertex l, identifying r_1 and r_2 into a single vertex r, and choosing (l, r) as the new terminal pair; and $G_1 \bullet G_2$ is obtained from the union of G_1 and G_2 by identifying r_1 and l_2 into a single vertex, and choosing (l_1, r_2) as the new terminal pair. For a two-terminal series-parallel graph G, we let $G^{<n>}$ denote the series composition of n copies of G, and let $G_{<n>}$ denote the parallel composition of n copies of G. For example, the 5-cycle C_5 with non-adjacent terminals is denoted by $(K_2 \bullet K_2)//(K_2 \bullet K_2 \bullet K_2)$, or alternatively $(K_2)^{<2>}//(K_2)^{<3>}$.

Lemma 6 *For all d and $k \geq 3$, there is a series-parallel graph of odd girth k such that removing the edges of a subgraph of maximum degree d cannot destroy every k-cycle.*

Proof. Assume first that $k = 3$. Construct graph G_3 as follows. Start with a complete bipartite graph $K_{2,d+1}$ and then for every edge, join its ends by $2d+1$ disjoint paths of length 2. In our notation, $G_3 = [((K_2 \bullet K_2)_{<2d+1>}//K_2)^{<2>}]_{<d+1>}$. Removal of the edges of a subgraph of maximum degree d from G_3 must leave at least one triangle.

Assume second that $k \geq 5$. Let graph $S_a = ((K_2 \bullet K_2)_{<2d+1>})^{<a>}$. (Thus S_a has diameter $2a$.) Then let graph $G_k = (S_{\lfloor k/4 \rfloor} \bullet K_2)_{<d+1>}//S_{\lfloor (k+2)/4 \rfloor}$. See Figure 3. Clearly G_k has odd girth k, and removal of the edges of a subgraph of maximum degree d from G_k must leave at least one k-cycle. $\square$

From this it follows:

Theorem 7 *For all d and $k \geq 3$, the maximum d-defective fractional chromatic number of a series-parallel graph of odd girth k is $2k/(k-1)$.*

PROOF. The upper bound follows from Theorem 3. The lower bound follows from the above construction. $\square$

In particular, this theorem shows that for every d there is a series-parallel graph whose d-defective fractional and circular chromatic numbers are both 3. This answers Klostermeyer's question [9] whether every series-parallel graph has a 2-defective $(5,2)$-circular coloring.

The above construction carries over partially to d-defective circular chromatic number. In particular:

Theorem 8 *For all d, the maximum d-defective circular chromatic number of a triangle-free series-parallel graph is $8/3$.*

PROOF. The upper bound follows from the result (Theorem 1.1) of Hell and Zhu [5].

For the lower bound, let graph $F_d = [(K_2 \bullet K_2)_{<2d+1>} \bullet K_2]_{<d+1>}//(K_2 \bullet K_2)_{<2d+1>}$, graph $G_d = [K_2 \bullet (K_2 \bullet K_2)_{<2d+1>}]_{<d+1>}//(K_2 \bullet K_2)_{<2d+1>}$, and graph $H_d = (F_d \bullet F_d)//G_d$. The graph H_1 is shown in Figure 4. Clearly H_d is triangle-free. It can readily be shown that if we remove the edges of a subgraph of maximum degree d from H_d, the remaining graph still contains a copy of the graph of Figure 1. Hence H_d has d-defective circular chromatic number $8/3$. $\square$

The above construction also provides a counterexample to Klostermeyer's claimed result [9] that every triangle-free series-parallel graph has a 2-defective $(5,2)$-circular coloring.

By starting with the extremal graphs constructed by Pan and Zhu [13], one can similarly show that the maximum d-defective circular chromatic number of a series-parallel graph of **odd girth** k is at least the maximum circular chromatic of a series-parallel graph of **girth** k. But there does not seem any reason to believe that the values are equal, since the question of the maximum circular chromatic of a series-parallel graph of **odd girth** k is unresolved.

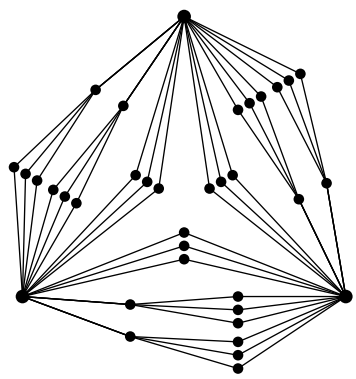

Figure 4: The graph H_1 whose 1-defective circular chromatic number is $8/3$

4 Related Questions

Recall that series-parallel graphs are also the partial 2-trees (see for example [4] for a discussion of this and the requisite definitions). So it is natural to consider partial k-trees in general. For example, Chlebíková [2] showed that: for $k \geq 3$, every triangle-free partial k-tree has chromatic number at most k. So one question is whether this is best possible? Also what happens for fractional/circular coloring and/or higher girth/odd girth?

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